

ECEN 620

Homework 1

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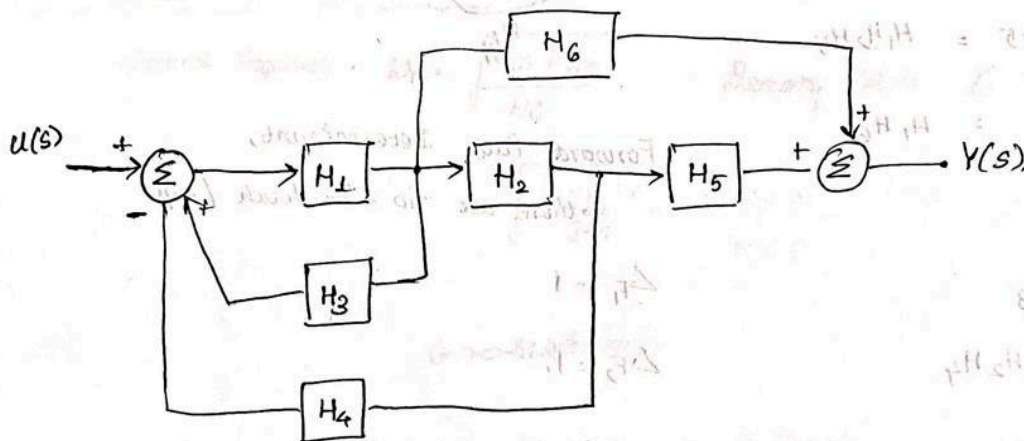
ECEN : 620

NETWORK THEORY.

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1. Use Mason's rule to derive the transfer function $G(s) = Y(s)/U(s)$ of the system below, Clearly state the forward path gains, loop gains, system determinant and forward path determinant.



Solution:

Mason's Gain Formula is given by,

$$\frac{C(s)}{R(s)} = \sum_{k=1}^n \frac{P_k \cdot \Delta_k}{\Delta}$$

n = no. of forward path

P_k = Forward Path gain of k^{th} forward path

Δ_k = Associated Path Factor (cofactor)

Δ = Determinant of SPG. {Signal Flow Graph}

Δ = Determinant of SPG,

$= 1 - [\text{Sum of gains of all the individual loops}]$

$+ [\text{Sum of product of gains of all possible combinations of two non-touching loops}]$

$- [\text{Sum of products of gain of all possible combinations of three Non-touching loops}] + \dots$

Let's consider a 2nd order type 2 charge pump PLL.

$$K_{PD} = 20 \mu A / (2\pi) \text{ radians}$$

$$K_{VCO} = 2\pi * (10 \text{ GHz}) / V.$$

$$N = 16.$$

Find the specs for the loop filter such that.

$$\omega_n = 2\pi * 10 \text{ MHz for } \zeta = 1, 0.2, 5.$$

for Second Order Type 2 PLL:

$$\text{Natural frequency } \omega_n = \sqrt{\frac{K_{PD} K_{VCO}}{N C}}$$

$$\text{Damping factor } \zeta = \frac{\omega_n R C}{2}$$

$$2\pi * 10 * 10^6 \text{ Hz} = \sqrt{\frac{20 * 10^{-6} A}{2\pi} * \frac{2\pi * 10 * 10^6}{16 * C}}$$

~~$$C = 3.166 \text{ pF}$$~~

$$C = \frac{K_{PD} K_{VCO}}{N \omega_n^2} = 3.166 \text{ pF}$$

$$\boxed{C = 3.166 \text{ pF}}$$

R values:-

depends on the damping factor,

$$\zeta = 1,$$

$$\zeta = \frac{\omega_n R C}{2}$$

$$R = \frac{2 \zeta}{\omega_n * C} = \frac{2 * 1}{2\pi * 10 * 10^6 \text{ MHz}}$$

~~$$\omega_n = \frac{2 \zeta}{R C} = \frac{2 * 1}{R C}$$~~

$$\boxed{R = 10.05 \text{ k}\Omega}$$

$$\zeta = 0.2$$

$$R = \frac{2 * 0.2}{2\pi * 10 * 10^6 \text{ MHz}}$$

$$3.166 \text{ pF} * 2\pi * 10 * 10^6 \text{ MHz}$$

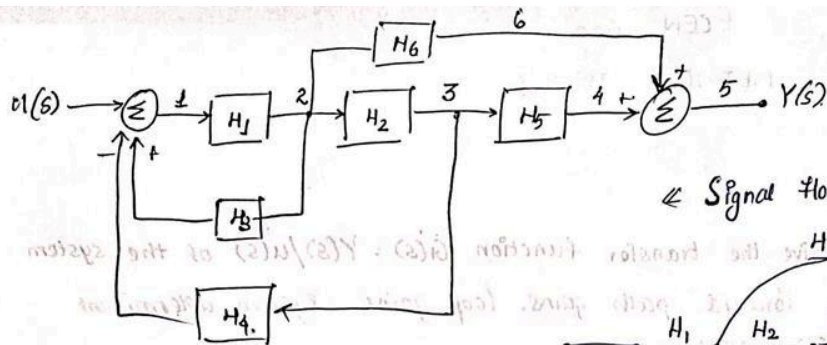
$$\boxed{R = 2.01 \text{ k}\Omega}$$

$$\zeta = 5$$

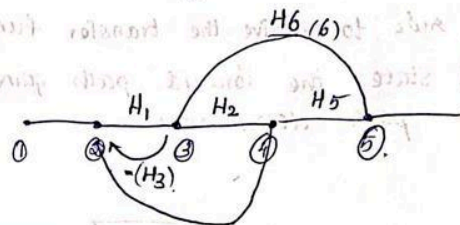
$$R = \frac{2 * 5}{2\pi * 10 * 10^6 \text{ MHz}}$$

$$3.166 \text{ pF} * 2\pi * 10 * 10^6 \text{ MHz}$$

$$\boxed{R = 50.26 \text{ k}\Omega}$$



« Signal Flow Graph »



Forward Path Gain,

$$G_1 = 12345 = H_1 H_2 H_5$$

$$G_2 = 1265 = H_1 H_6$$

Loop Gain

$$L_1 = H_1 H_3$$

$$L_2 = -H_1 H_2 H_4$$

Forward Path Determinant,

→ there are no Non Touch Loop

$$\Delta F_1 = 1$$

$$\Delta F_2 = 1.$$

System Determinant,

$$\Delta = 1 - (L_1 + L_2)$$

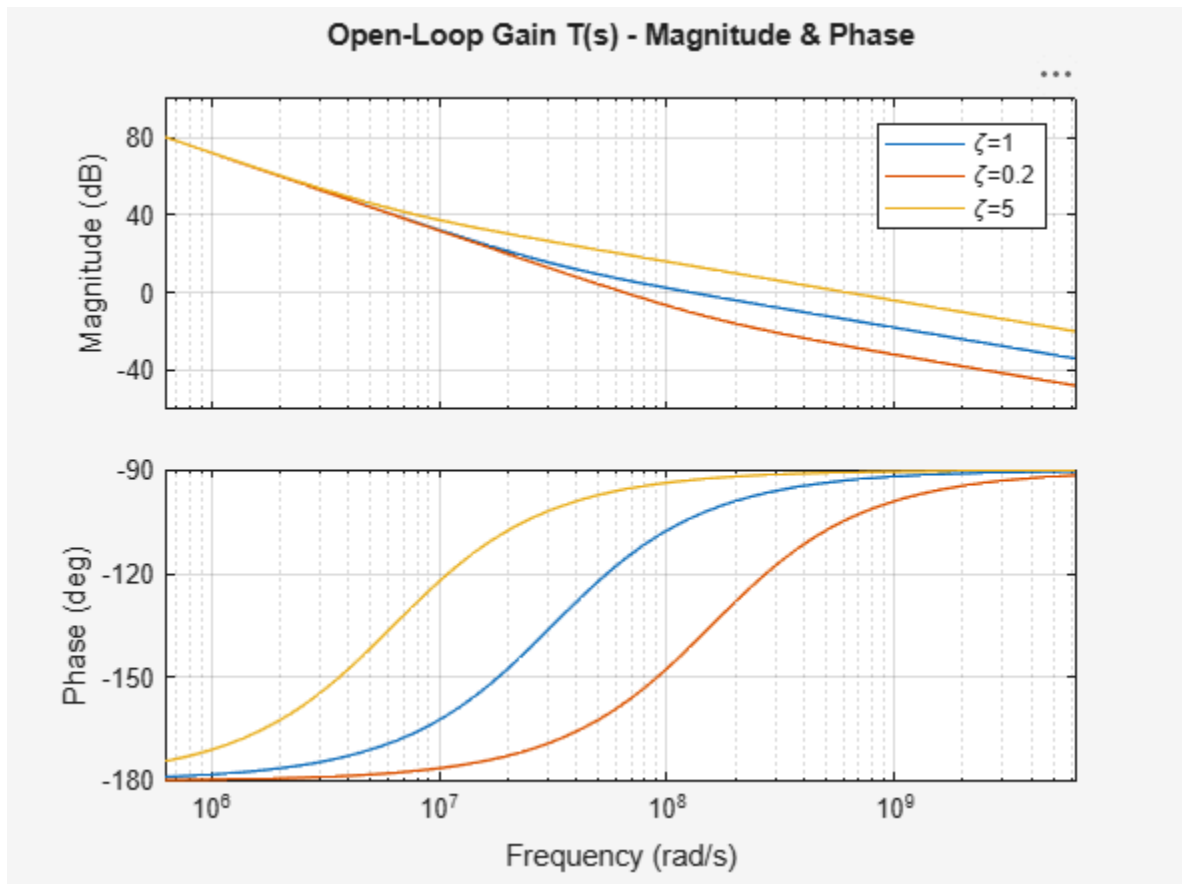
$$= 1 - (H_1 H_3 - H_1 H_2 H_4)$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{P_1 \Delta_1 + P_2 \Delta_2}{\Delta}$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{H_1 H_2 H_5 + H_1 H_6}{1 - (H_1 H_3 - H_1 H_2 H_4)}$$

Implement a linear macromodel (MATLAB, etc.) for the closed-loop system:

- Plot the open-loop gain magnitude and phase, and find the phase margin. For both the magnitude and phase plots, overlay all the curves for the different ζ values on one clearly labeled plot.



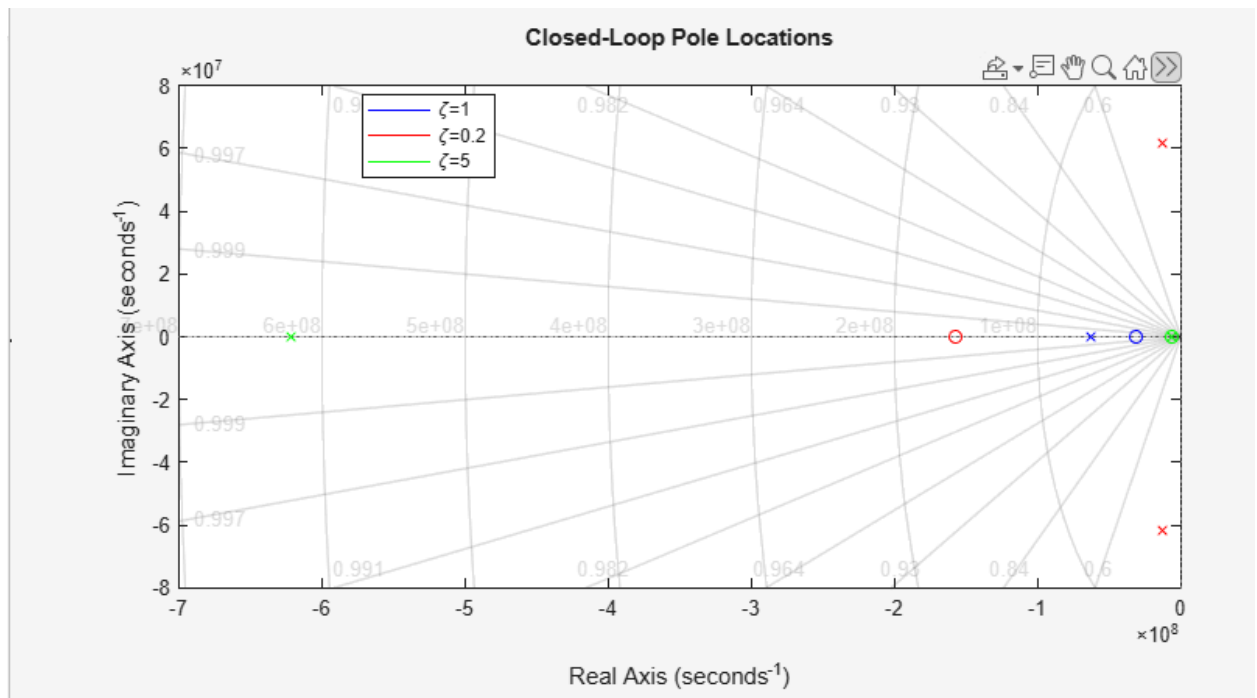
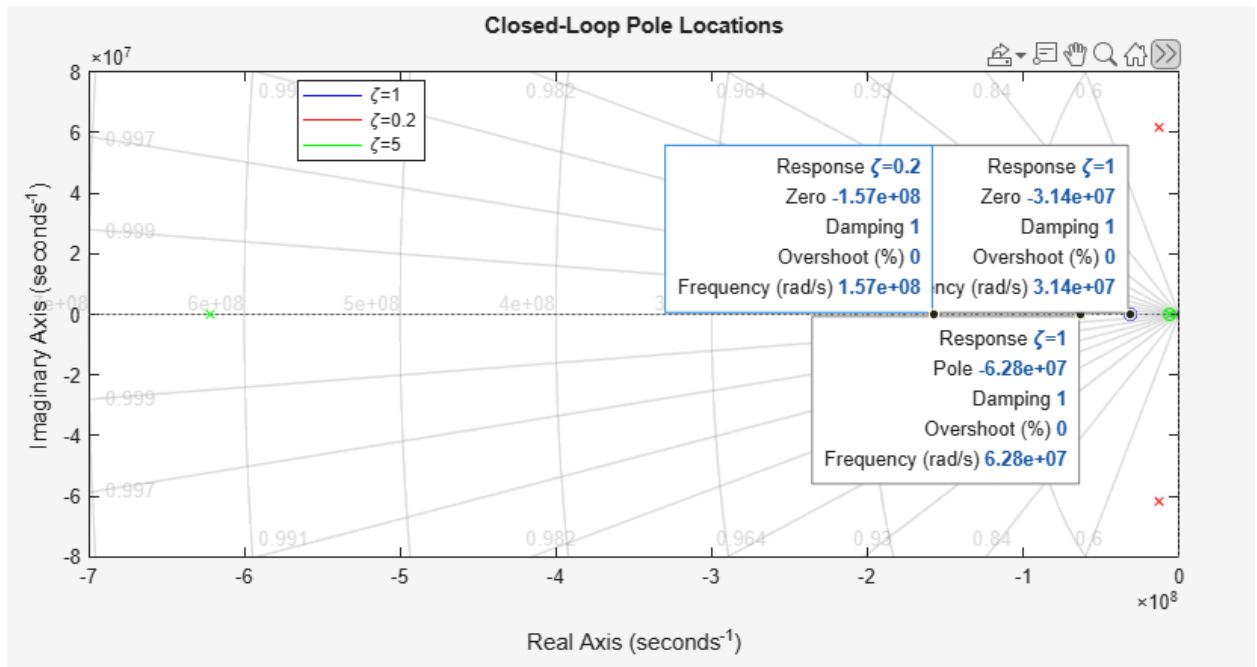
ζ Phase Margin

1.00 76.35°

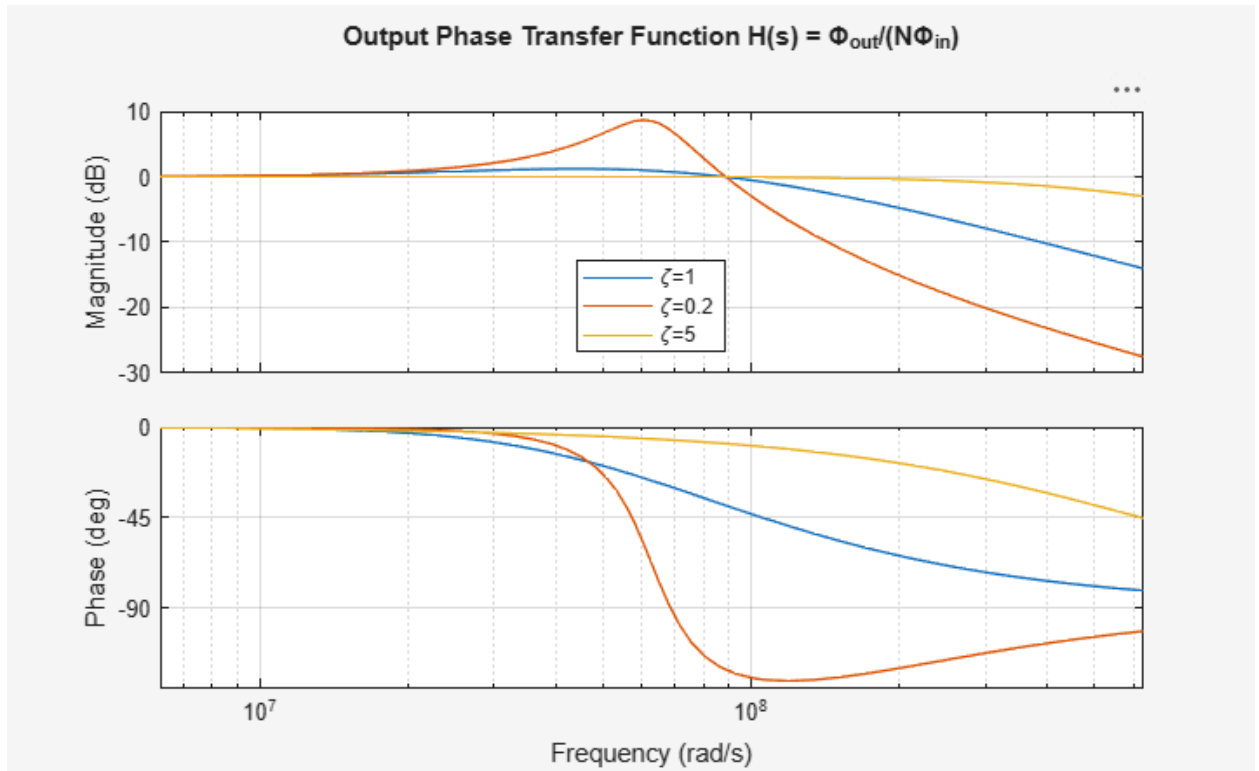
0.20 22.60°

5.00 89.43°

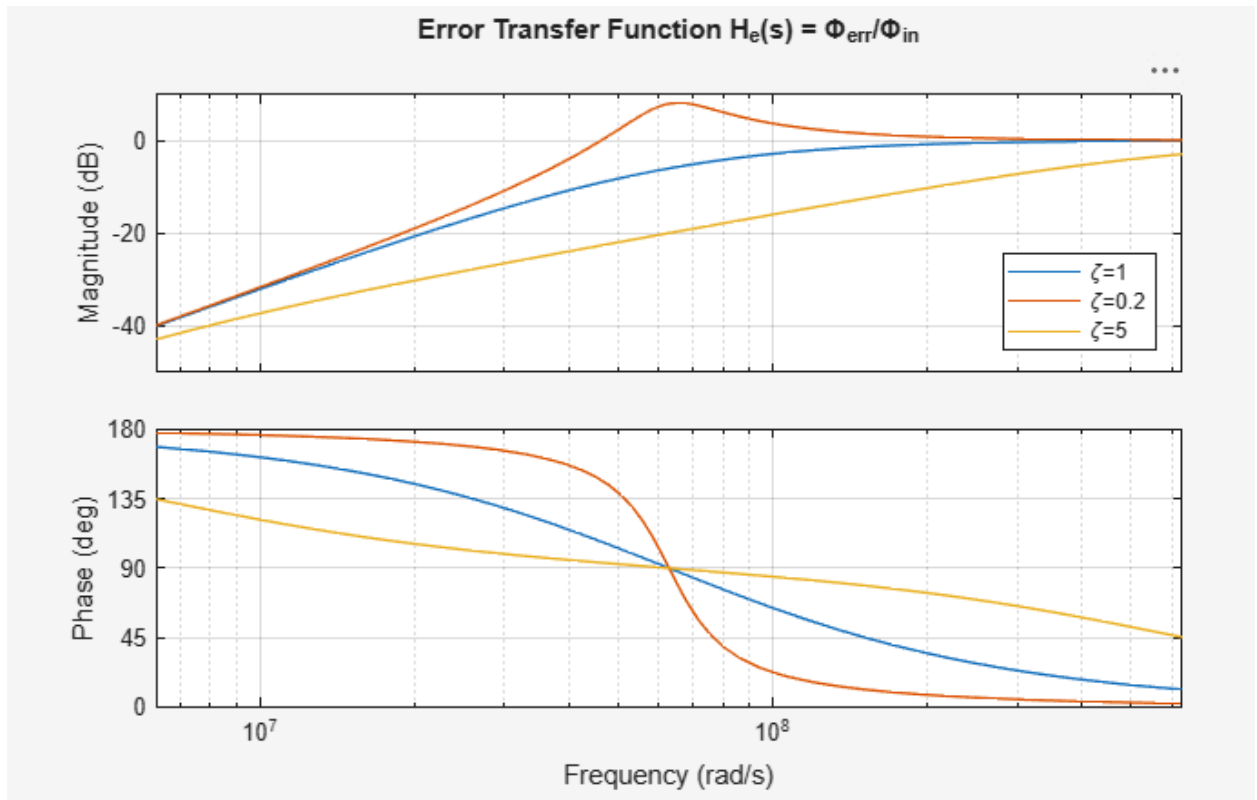
- Give the closed-loop pole values for $\zeta = 1, 0.2, 5$.



- Plot the output phase transfer function (magnitude and phase) and error transfer function (magnitude and phase) from $0.1\omega_n$ to $10\omega_n$. For both the magnitude and phase plots, overlay all the curves for the different ζ values on one clearly labeled plot.



- Plot the output phase and phase error transient response to a unit phase step. For both plots, overlay all the curves for the different ζ values on one clearly labeled plot.



- Using the information obtained from the frequency-domain and transient phase step plots, comment on the system behavior as ζ changes.

$\zeta = 0.2$ (Underdamped)

- Phase Margin: $\approx 23^\circ$ (low)
- Closed-loop poles: Complex, close to the j-axis
- Frequency-domain behavior: Resonant peak in $H(s)$ and $H_e(s)$ near ω_n
- Step response: Fast rise, but with overshoot and ringing
- Settling: Slow (decay set by small real part of poles)

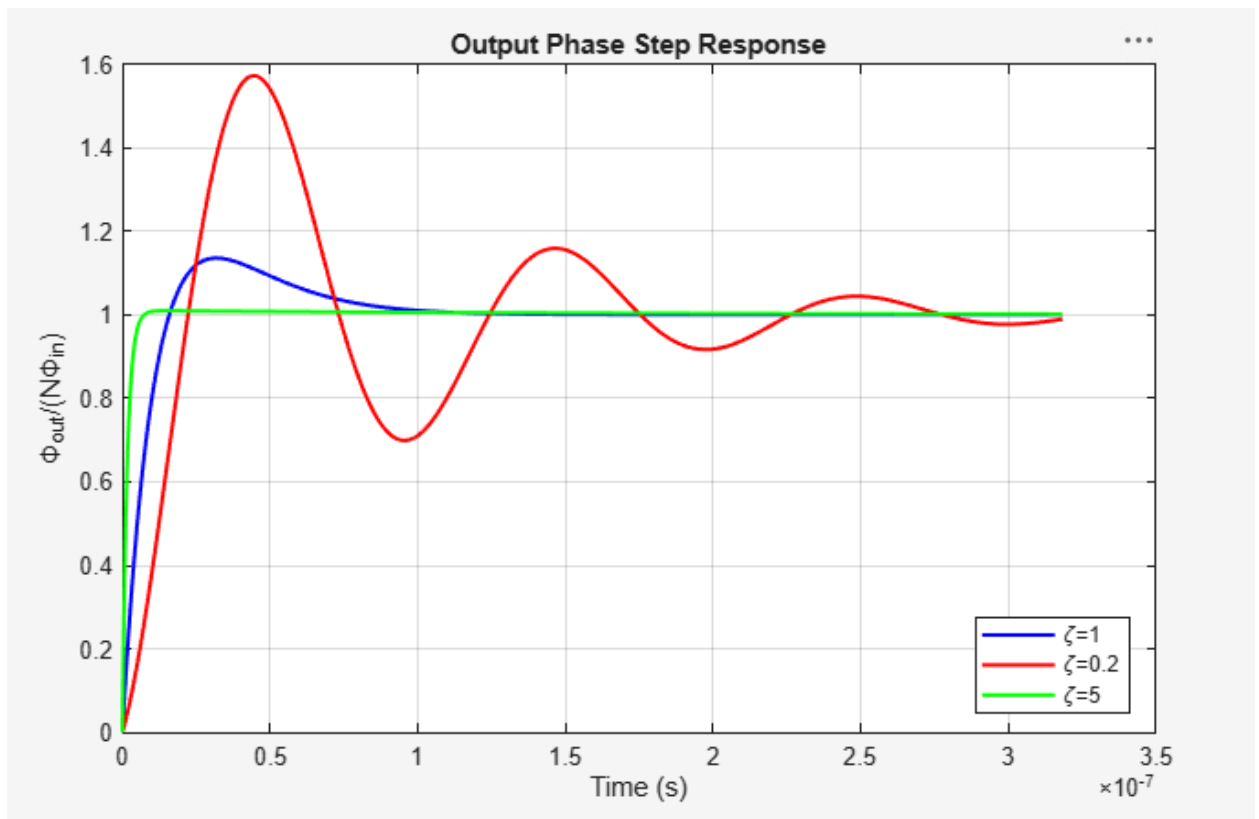
$\zeta = 1$ (Critically Damped)

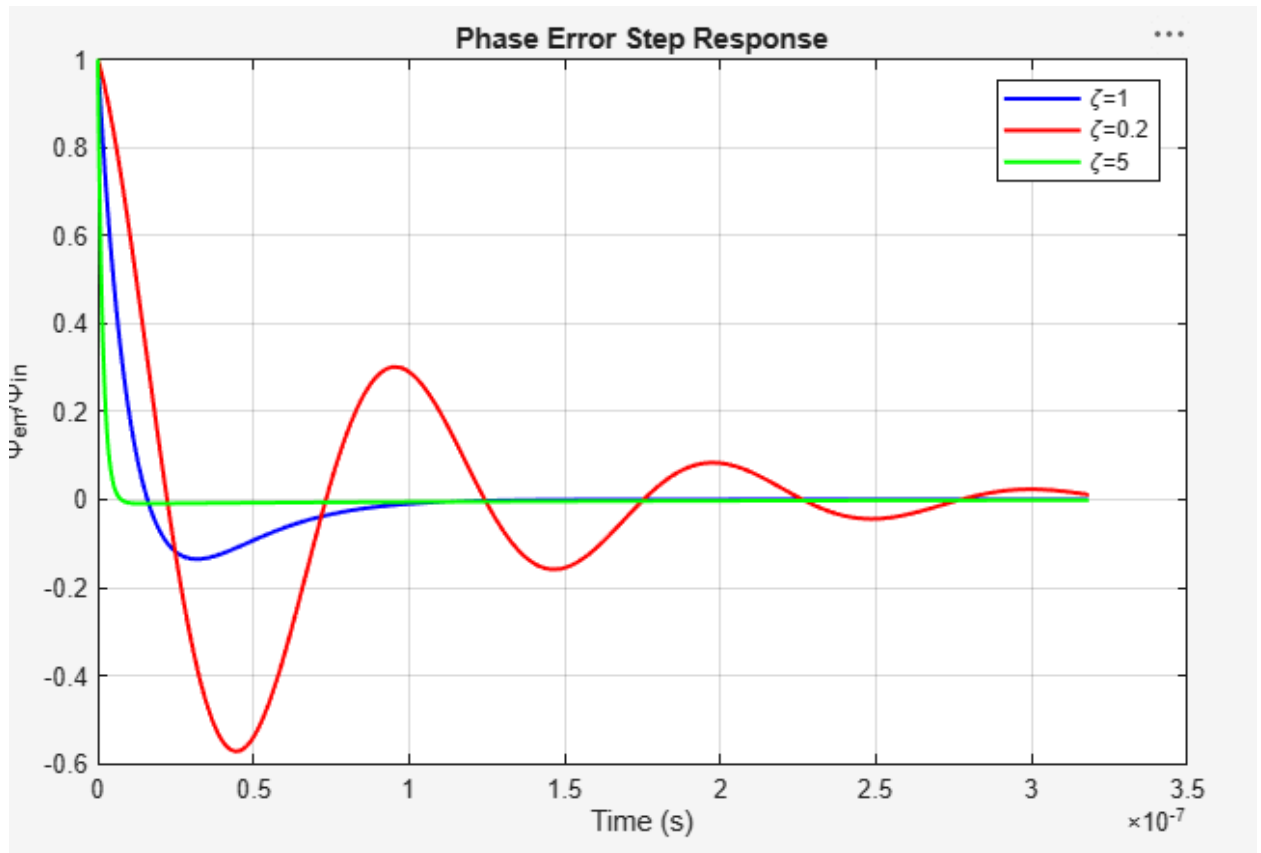
- Phase Margin: $\approx 76^\circ$ (moderate)
- Closed-loop poles: Real, double pole
- Frequency-domain behavior: Smooth roll-off, no peaking
- Step response: Fast rise, no overshoot
- Settling: Fastest overall

$\zeta = 5$ (Overdamped)

- Phase Margin: $\approx 89^\circ$ (high)
- Closed-loop poles: Real, widely separated
- Frequency-domain behavior: Extended low-pass shape, no peaking
- Step response: No overshoot

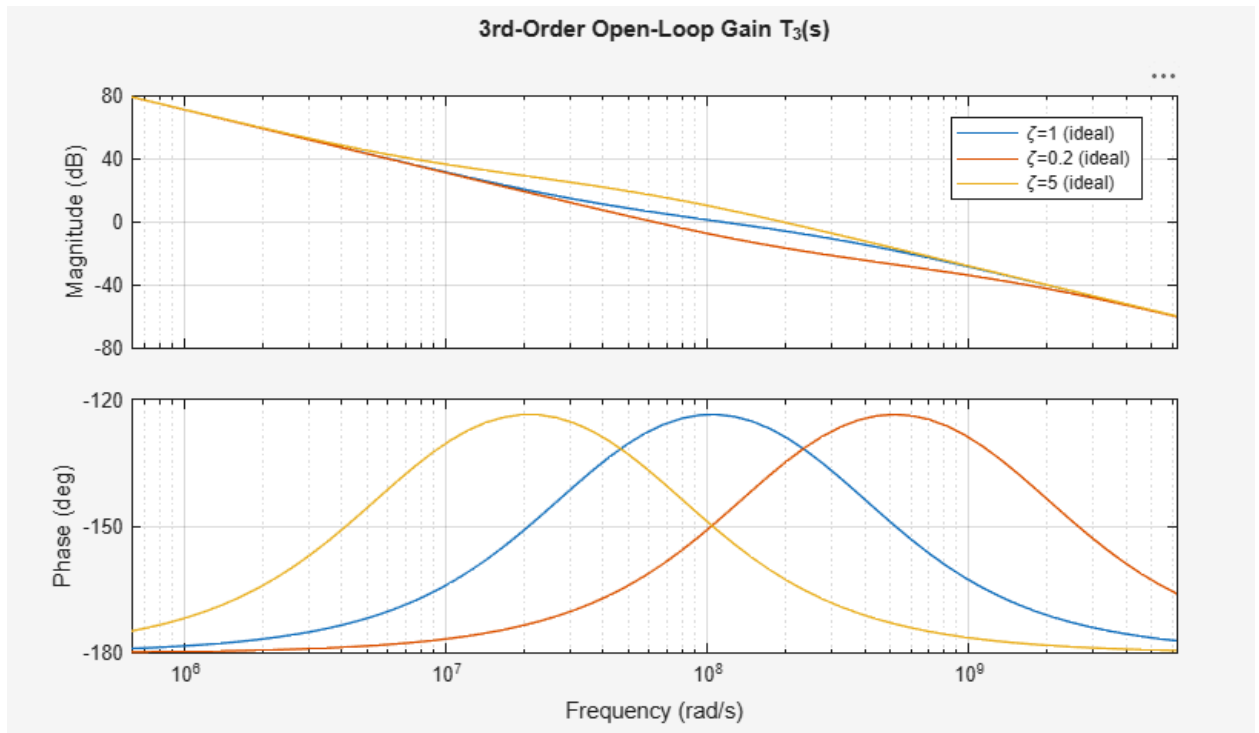
- Settling: Slow (dragged out by the far-separated slow pole)



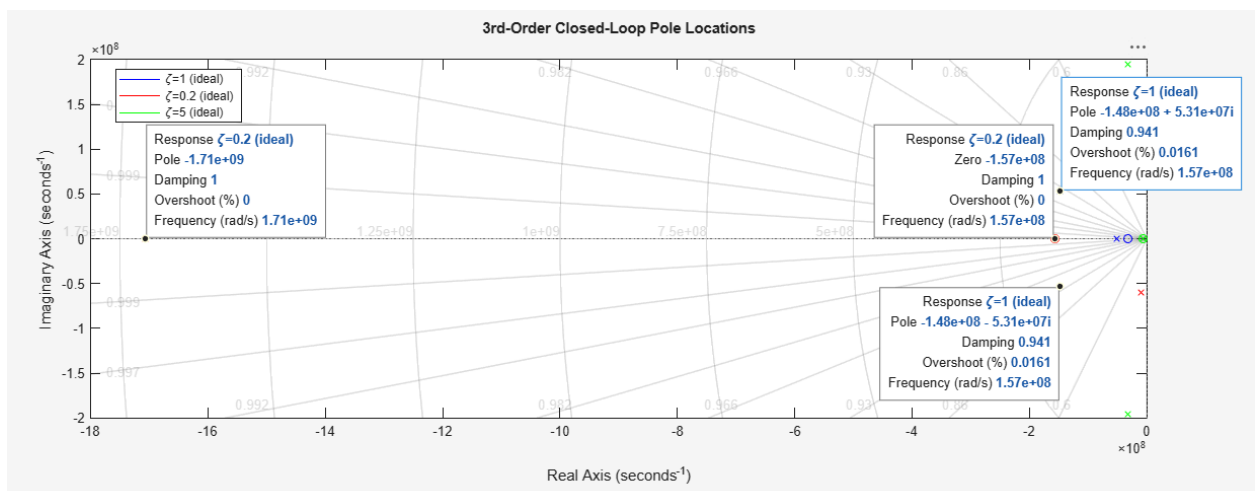


3. Now let's consider the more realistic case where we have an additional filtering capacitor C_2 in parallel to the main loop filter. Set $C_2=0.1 \cdot C_1$ and, without modifying the R and C_1 values from #1, repeat the previous plots

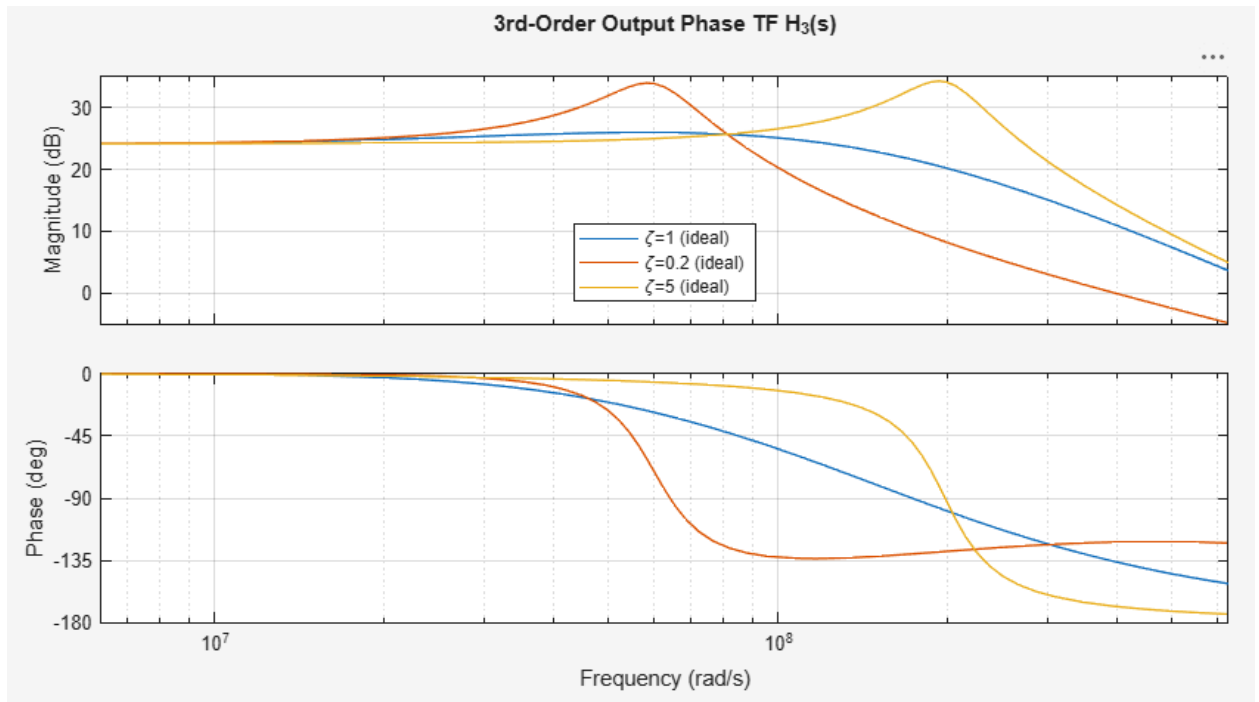
- Plot the open-loop gain magnitude and phase and find the phase margin. For both the magnitude and phase plots, overlay all the curves for the different ideal 2nd order ζ values on one clearly labeled plot.



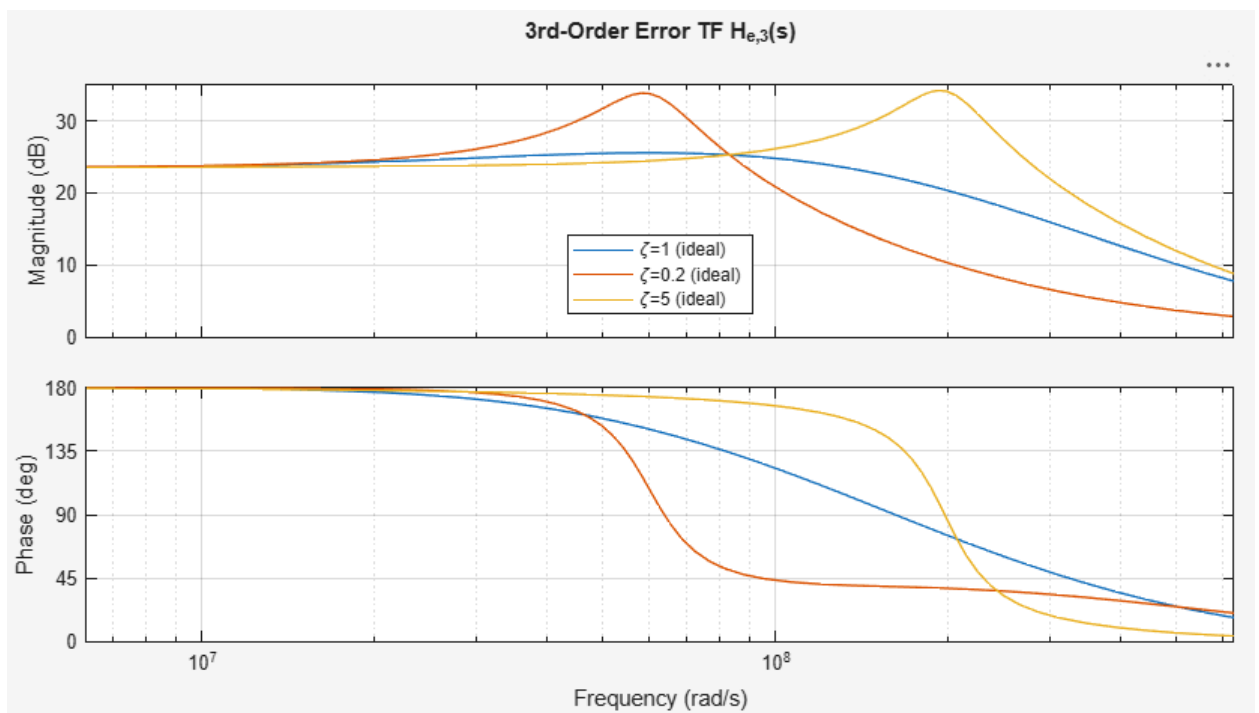
- Give the closed-loop pole values for ideal 2nd order $\zeta = 1, 0.2, 5$.



- Plot the output phase transfer function (both magnitude and phase) and error transfer function (magnitude and phase) from $0.1\omega_\square$ to $10\omega_\square$. For both the magnitude and phase plots, overlay all the curves for the different ideal 2nd order ζ values on one clearly labeled plot.



- Plot the output phase and phase error transient response to a unit phase step. For both of these plots, overlay all the curves for the different ideal 2nd order ζ values on one clearly labeled plot.



- Using the information obtained from the frequency-domain and transient phase step plots, comment on the system behavior as the ideal 2nd order ζ changes. How is this more realistic 3rd order system different from the ideal 2nd order system of #1?

C2 adds a pole at $\omega_p = 1/(R \cdot C_1) = 11 \cdot \omega_z$ on top of the existing zero. Because $\omega_z = \omega_n/(2\zeta)$, high- ζ designs have a low ω_z , so ω_p lands close to ω_n — this is why the $\zeta = 5$ case loses the most phase margin, with its closed-loop poles flipping from two real poles to an underdamped complex pair. The $\zeta = 0.2$ design keeps ω_z high, so ω_p is pushed far out and the response is barely changed from the ideal 2nd-order case. The $\zeta = 1$ case also degrades, from critically-damped to mildly underdamped.